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Evolving FAIR Image Analysis in Galaxy for Cross-domain and AI-ready Applications

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1 | Introduction

The growing integration of image-based technologies across life sciences, environmental research, and other domains has intensified the need for interoperable, reproducible, and FAIR-compliant image analysis infrastructures. The ELIXIR BioHackathon Europe 2025 provided a collaborative environment to advance these goals within the Galaxy ecosystem (Galaxy Community, 2024).

Previous projects at the ELIXIR BioHackathon Europe focused on community development and establishing conventions (Kostrykin et al., 2024) or implementing FAIR workflows and training materials (Kostrykin et al., 2025) for image analysis using Galaxy. Building on the achievements of the previous years, Project 9 at the ELIXIR BioHackathon Europe 2025 was entitled *"Evolving FAIR Image Analysis in Galaxy for Cross-domain and AI-ready Applications"* and aimed to strengthen Galaxy's capabilities for bioimage analysis while enhancing cross-domain interoperability and reproducibility. The project addressed three major challenges:

1. Improving semantic annotation for image analysis resources.
2. Introducing content-based reproducibility validation mechanisms.
3. Streamlining tools and training materials for accessible bioimage analysis.

To achieve these goals, the project involved participants with heterogeneous backgrounds from different institutions, including ELIXIR, Euro-BioImaging, and the ISCC Foundation, joining both on-site and online. The goals of the project were also closely aligned with the OSCARS-FIESTA and OSCARS-BIOCODES projects.

By integrating ontology development with standard-based validation tools, the project contributed to a FAIR-aligned, cross-domain image analysis framework within Galaxy.

2 | Implementation

Our work was structured into three coordinated work packages (WPs) as described below. WP1 focused on cross-domain discovery and interoperability (Section 2.1), WP2 on integrating content-based fingerprints to enhance reproducibility and AI-readiness (Section 2.2), and WP3 on streamlining bioimage analysis in general, by improving tools, training materials, and the usability thereof (Section 2.3).

2.1 | WP1: Cross-domain Support Improvements

WP1 focused on semantic interoperability through improvements to the EDAM Bioimaging ontology. The EDAM Bioimaging ontology extends the core EDAM ontology (Ison et al., 2013)

with terms specific to bioimage data and analysis. However, this ontology extension is still under development. In the future, it will enable fine-grained annotation of tools, workflows, and tutorials within our community for specific analysis tasks. The BioHackathon team proposed extensions to the EDAM Bioimaging ontology to better align with the image analysis tasks supported by Galaxy. Furthermore, we curated a specialized vocabulary that reflects both the latest EDAM Bioimaging version and our specific recommendations (Section 3.1), which was then used to annotate the image analysis tutorials on the Galaxy Training Network (Hiltemann et al., 2023). This enhances the discoverability of the tutorials in the short term while ensuring seamless compatibility with evolving standards.

Through this work package, semantic annotation became a central mechanism to enable cross-domain discovery, interoperability, and alignment with evolving community standards.

2.2 | WP2: Integration of the ISCC ISO Standard in Galaxy

WP2 introduced content-based validation mechanisms by integrating the International Standard Content Code (ISCC) into Galaxy. To support reproducibility, the ISCC-SUM tool suite was integrated into Galaxy. This generates content-based fingerprints to validate final or intermediate results by implementing the International Standard Content Code (ISCC). ISCC-SUM is file-type agnostic and not limited to bioimaging data; it can be extended to various heterogeneous data types, such as omics data. This content-based validation complements Galaxy's workflow provenance tracing, providing format-independent reproducibility assurance aligned with FAIR principles.

Three Galaxy tools were implemented: (BMCV/galaxy-image-analysis/pull/159, BMCV/galaxy-image-analysis/pull/162)

1. A tool to compute ISCC codes for datasets.
2. A tool to compare ISCC codes based on a configurable similarity threshold.
3. A tool to filter datasets according to ISCC similarity constraints.

These tools can be used to incorporate reproducibility-assuring checks as workflow steps directly into analysis pipelines in Galaxy.

The integration of the ISCC-SUM tool suite into Galaxy enhances the AI-readiness of the platform. In this context, ISCC codes have several advantages. One advantage is that ISCC codes enable deduplication and dataset cleaning, which is important for removing redundant training data to prevent bias from over-represented samples. Another advantage is that they allow grouping similar (or dissimilar) data. This can be used, for example, during model training, to select test sets that truly evaluate the generalization of the model. When using pre-trained models, this can also be used to determine whether the image data used in an analysis is too different from the training data (out-of-distribution data), and thus to decide whether a tool or model is appropriate for that image data. Moreover, ISCC codes can also be used to verify data integrity, thus generally enhancing the trustworthiness of analyses.

Citation Typing Ontology annotation

You can use CiTO annotations, as explained in this BioHackathon Europe 2021 write up and this CiTO Pilot. Using this template, you can cite an article and indicate *why* you cite that article, for instance DisGeNET-RDF (Queralt-Rosinach et al., 2016).

The syntax in Markdown is as follows: a single intention annotation looks like `[@usesMethodIn:Krewinkel2017]`; two or more intentions are separated with colons, like `[@extends:discusses:Nielsen2017Scholia]`. When you cite two different articles, you use this syntax: `[@citesAsDataSource:Ammar2022ETL; @citesAsDataSource:Arend2022BioHackEU22]`.

Possible CiTO typing annotation include:

- citesAsDataSource: when you point the reader to a source of data which may explain a claim
- usesDataFrom: when you reuse somehow (and elaborate on) the data in the cited entity
- usesMethodIn
- citesAsAuthority
- citesAsEvidence
- citesAsPotentialSolution
- citesAsRecommendedReading
- citesAsRelated
- citesAsSourceDocument
- citesForInformation
- confirms
- documents
- providesDataFor
- obtainsSupportFrom
- discusses
- extends
- agreesWith
- disagreesWith
- updates
- citation: generic citation

Results

Discussion

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Acknowledgements

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Author contributions